

Group B Practical 1

// 1. Implement a program to calculate precision and recall for sample input.

// Package CalculatePrecisionandRecall;

import java.util.*;

public class CalculatePrecisionandRecall {

public static void main(String[] args) {

// Sample documents and queries

// A document in IR is the basic unit of retrieval

// Document: "Information retrieval (IR) is the process of obtaining information"

Map<String, String> documents = new HashMap<>();

documents.put("Document1", "Information retrieval involves various techniques for extracting information");

documents.put("Document2", "Text analysis process for information retrieval");

Map<String, String> queries = new HashMap<>();

queries.put("Query1", "Information retrieval techniques obtaining information from a database");

queries.put("Query2", "text analysis process");

// Relevance judgments (manually annotated for simplicity)

Map<String, List<String>> relevanceJudgments = new HashMap<>();

relevanceJudgments.put("Query1", Arrays.asList("Document1"));

relevanceJudgments.put("Query2", Arrays.asList("Document2"));

// Calculate precision and recall for each query

for (String queryId : queries.keySet()) {

String query = queries.get(queryId);

List<String> relevantDocuments = relevanceJudgments.getOrDefault(queryId, new ArrayList<>());

// Tokenize the query and document

Set<String> queryTokens = new HashSet<>(Arrays.asList(query.split(" ")));

Set<String> documentTokens = new

HashSet<>(Arrays.asList(documents.get(queryId.replace("Query", "Document")).split(" ")));

// Calculate precision and recall

double precision = calculatePrecision(queryTokens, documentTokens);

double recall = calculateRecall(queryTokens, documentTokens);

```

        System.out.println("Query: " + query);
        System.out.println("Relevant Documents: " + relevantDocuments);
        System.out.println("Precision: " + precision);
        System.out.println("Recall: " + recall);
        System.out.println();
    }
}

private static double calculatePrecision(Set<String> queryTokens, Set<String>
documentTokens) {
    Set<String> intersection = new HashSet<>(queryTokens);
    intersection.retainAll(documentTokens);
    if (queryTokens.isEmpty()) {
        return 0.0;
    }
    return (double) intersection.size() / queryTokens.size();
}

private static double calculateRecall(Set<String> queryTokens, Set<String>
documentTokens) {
    Set<String> intersection = new HashSet<>(queryTokens);
    intersection.retainAll(documentTokens);
    if (documentTokens.isEmpty()) {
        return 0.0;
    }
    return (double) intersection.size() / documentTokens.size();
}
}

```

Output

Query: Information retrieval techniques obtaining information from a database
Relevant Documents: [Document1]
Precision: 0.5
Recall: 0.5

Query: text analysis process
Relevant Documents: [Document2]
Precision: 0.6666666666666666
Recall: 0.3333333333333333

=== Code Execution Successful ===